Addendum:
Howdy!

As you are reviewing this I would like to remind you that I discussed with Dr.Righetti that my team dissolved. and presenting this solo. Due to the change in expertise, there was a shift in the project implementation, though the core of DIP remains intact. Sorry for the inconvenience, and hope this project is to your liking.

1.Introduction

The project tests the use of DIP techniques in the context of unsupervised anomaly detection for eggshell crack detection. The goal is to analyze how different techniques affect performance. These are for this project:

- Spatial Cropping
- CLAHE
- Morphology
- Edge Detection
- Combined (Crop+CLAHE+Edge)

All of these images were greyed,blurred, and resized to meet the input requirements of RestNet50 and provide a uniform experience.

These images were fed to a pretrained model, RestNet50, as a deep feature extractor with different types of filters as mentioned above. This is a change from the original proposal, as it had been set to employ anomalib, which focuses on anomaly detection. However, anomalib is quite unstable, especially with Google Collab which is where the experiments were run. The main issue was version incompatibilities between Collab, anomalib and some changes it underwent a couple of weeks ago, thus the change was necessary. I opted for a different model that can perform the same task. The main effect of this change on the project is the accuracy on data, as this model is worse. However, this is a DIP class, thus the model precision is not the main concern irrelevant when compared with what we want to see the effects of the processing has on the data.

Thus, even if different, the project should hit the focus of the class, as well as expand more on the topic of image processing. We lose overall state performance, but learn more about the performance effect the preprocessing has on the results.

2.Dataset

To run the unsupervised regime, images from different online sets were obtained. There currently are1971 images on the set:

- 1511 images uncracked eggs for training purposes
- 377 uncracked for testing purposes
- 91 cracked eggs for testing purposes

From the following public sources we assembled our final repository for the experiment:

- https://www.kaggle.com/datasets/abdullahkhanuet22/eggs-images-classification-damaged-or-not
- https://www.kaggle.com/datasets/frankpereny/broken-eggs/data
- https://figshare.com/articles/dataset/Dataset_for_real-time_crack_detection_on_chicken_eggs/21568425
- https://data.mendeley.com/datasets/mdty358x8m/1?utm

A curated combination of the above images can be found below:
:https://drive.google.com/drive/folders/1wl3PwAU6EdhfnH6krWhbq9vbq6v-_mr1?usp=drive_link
https://drive.google.com/drive/folders/1eof9xFmMYmJZj0KK7JNRbiAcuOXCm-sP?usp=drive_link

To run the notebook you must have access to both drives, as the code pulls them of the repository then applies the preprocessing

Issues

As shown in the image bases used, there is a lack of properly cracked images on the internet that are decently clean to experiment with. Databases mainly focus on uncracked images. The images suffer from heavy background unfortunately, but we fix this with general interventions for cleaning it. We could test on cracked images that are truly clean, but for industrial purposes this is rarely the case, and adding some noise ensures the model truly learns rather than relying on a secondary effect, like the background or shape of the canister that holds it. On top of this, the final result is not meant to be used but rather shows the power of general preprocessing.

The images also presented different lighting, clutter, and small cracks, which the filters we designed help deal with, and at the core of the assignment.

3.Preprocessing Methods:

In this section, the methods are explained that were used to preprocess the data.

For the baseline, all images regardless of technique used received:

- Grayscaling
- Gaussian Blur
- Resizing resolution

The use of this is necessary as it simplifies the learning that has to be performed, and cleans the images of unnecessary details. Eggs come from white to brown, some even have specks, but these do not contribute to determining if there is a crack. Thus we cancel that with grayscale and focus on learning the actual cracks or intensities, rather than color based features.

Gaussian blur was applied to lower high-noise details, and finer details that would hurt the learning process, and suppress the specks that are part of the egg detail. This operation removes finer details, but preserves large structures like cracks that actually impact the quality of the egg. A note to add: I went with a 3x3 filter after comparing it with 5x5. The 5x5 hurt the cracks a lot and dropped the score significantly, so we continued with 3x3.

Finally, all images were set to 256x256 as it is a base requirement for RestNet50 to run and ensure uniformity. Figure 1 illustrates an example of an image with the filter applied:

Figure 1: Egg with baseline filters

3.2 Tested Techniques

As mentioned before, several techniques were to see their effect on anomaly detection. They address background noise, clutter, illumination,contrast, and edge clarity.

Cropping

Spatial Cropping was used to remove the border region of the images aggressively, before resizing. Doing so removes noise from border features such as the walls of the pictures, and unnecessary information.

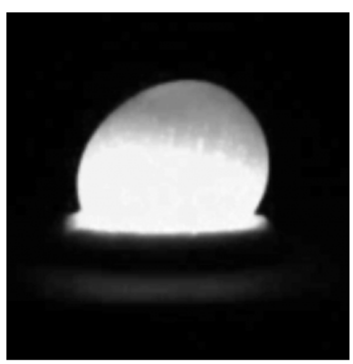

Figure 2: Cropped Image

Contrast Limited Adaptive Histogram Equalization (CLAHE)
CLAHE was used to locally enhance contrast for the image. It allows cracks to be more visible through histogram equalization and prevents the image from being too dark or bright, as well as preventing noise.

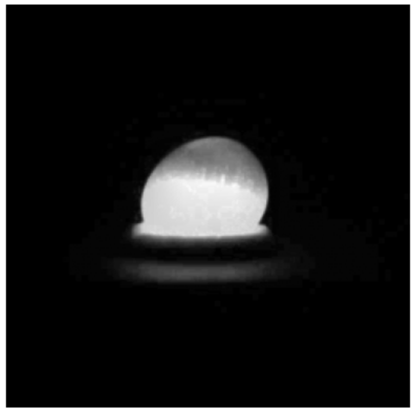

Figure 3: CLAHE image

Morphology
Morphology was introduced to accentuate features and see if the model could learn based on that. I directly chose to amplify the speckles by filling them with an ellipse, as they can be confused with cracks.

Figure 4: Morphology Image with opening

Edge Detection
Edge Detection features were extracted using the Sobel + Laplacian as cracks are sudden, allowing them to be more visible and easier to extract. This also follows for the shape of the egg should be (uncracked).

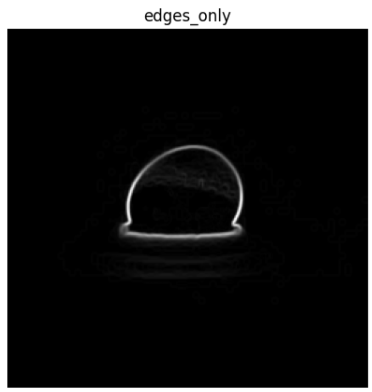

Figure 5: Edges

Combined

Finally, the final method combines Crop+CLAHE+Edge layered on top of each other to see how the multiple techniques help achieve accuracy once combined.

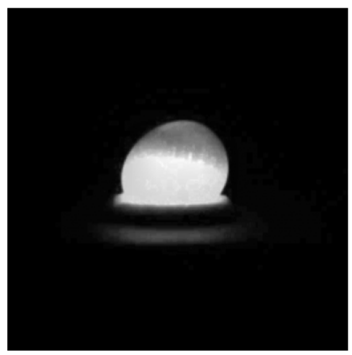

Figure 6: Combined

4. RestNet50

Following the preprocessing, the images were fed to a pretrained RestNet50 to extract their features and predict if they are normal eggs, or cracked. The model had its weights frozen during training to prevent the model itself from learning, thus the anomaly detection was the one that learned and the preprocessing is the only thing being affected and measured. The last layer was removed as it is used to predict RestNets own classification goals (horse,chair). Thus, we keep the information that can lead to classification and apply it to our case.

Isolation Forest was used to perform the detection on the cracked eggs. This works as we only trained on good eggs. The contamination rate was set (about how many of the eggs are in theory damaged) to .1 to consider for imperfection as eggs don't come perfect or the same, yet aren't cracked. This remained the same across all tests, and once again the main focus of this experiment is the improvement gain.

The implementation was also partly taken from another class that dealt with the topic. So many thank yous to Dr.Ji for helping with this.

4.1 Running It Yourself
I added a small section if you wish to run it. You have to open the provided notebook and activate the cells on order. FInally on the last cell with the training, change the final line

```
result = run_anomaly_detection(mode="morph", method="iforest",
contamination=0.1)
```

To your desired mode `["crop", "clahe", "edges_only", "combined","morph"]`

Also if you want to test the filters make a folder with 2 subfolders and an image of your choice. Locate the cell with test_process and modify your dataset_path so the name of the main folder and then subfolder. There is also a loaded version if you don't want to load it yourself.

```
test_preprocess(
    dataset_path="kachoow/kachoow2",
    modes=["crop", "clahe", "edges_only", "combined","morph"]
)
```

5.Results + Analysis

Processing Method	AUC-ROC	Accuracy	Crack Precision	Crack Recall	Crack f1
Cropped	0.8358	0.7582	0.5224	0.3846	0.4430
CLAHE	0.8773	0.7665	0.5556	0.3297	0.4138
Edge Detection	0.8291	0.7363	0.4359	0.1868	0.2615
Morphology (3x3)	0.8415	0.8022	0.6267	0.5165	0.5663
Morphology(5x5)	0.8477	0.8187	0.6866	0.5055	0.5823
Combined (No crop)	0.8598	0.7857	0.5942	0.4505	0.5125
Combined	0.8875	0.8324	0.6786	0.6264	0.6514

Table 1: Results of Experiments

Above we can see the results of the experiments that were ran for the classification. First the metrics used:

1. **AUC-ROC**
    a. Measures how well the anomaly detector does at classifying normal eggs, from cracked ones. Regardless of the case.If we pick an egg at random, how often it classifies it as cracked. In this scenario this is weaker since the majority of the eggs are cracked.
2. **Accuracy**
    a. Measures how many correct classifications were done from the expected results and the actual ones
3. **Crack Precision**
    a. Measures how many of the cracked images are actually cracked
4. **Crack Recall**
    a. Measures how many of the actual cracked eggs are correctly identified as cracked
5. **Crack F1**
    a. Balances Recall and Precision for a better score

For the evaluation, all the tests were performed as mentioned before. In addition, a second morphology test that used a different gaussian blur for its preprocessing, as well as an extra test for combining where no crop was performed for greater options, and to note qualities on the techniques.

From the results we can see the clear winner across all metrics is Combined, except for precision where morphology(5x5) beats it slightly. We can also see the worst performance comes from CLAHE alone, though crop and edge detection perform terrible as well. On top of that the results at classifying aren't terrific, since our data is noisy, and our model could be better.

First, we see that our techniques have a great effect on performance. For example cropping alone performs better than CLAHE and edge. This is probably because the model is trained with a large set of images, so it understands how to deal with illumination, and is not used to only having the edges. However, this doesn't explain why when combined with the other two it improves the model. For this we must consider that due to being trained in more features, the model is looking for those other features for its original classification. We are cutting off its data, and centering to only learn the egg. This is why cropping and reducing the background is so important. On top of that combining CLAHE and Edge Detection is great as it helps with accentuating the lines of eggs, and improving the detail of the image.

On top of that, we also note the importance of regulating the texture of the egg shell by using morphology. Morphology with a gaussian blur of (5x5) performs better than the (3x3) at correctly catching cracked eggs, especially its precision. The speckles we try to fill during morphology become more uniform and filled with a greater gaussian filter,so they don't appear as cracks. Our catches are more certain.

Overall, these findings show the importance of preprocessing, as well as the benefits different techniques can have on the metrics individually.

6.Conclusion

This project has investigated the impact of digital image processing techniques on unsupervised anomaly detection for eggshell crack detection. By isolating a model to focus on analyzing the effects the preprocessing has on its output, we show the importance of preprocessing on training a good model despite its data.

Among the tested methods, the best performance came from a pipeline using CLAHE, spatial cropping, and edge emphasis. The most important being cropping, as it limited the amount of background data the model took and directed it to learn from the target, the egg. Morphological processing also showed how it can handle uneven textures that the model could pick up, leading to finding cracks easier.

Though the results are not the best due to the model used, the data being noisy and hard to come by, and simple techniques, we can confidently say we can overcome issues like these with proper image processing techniques, and why no one focusing on machine learning can go without having experience in digital image processing.

Future work should focus on procuring better data, test more rigorous methods like segmentation and better models for detection. But as for this, we properly show the power of Digital Image processing.